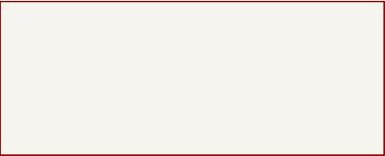


Ghost Fader

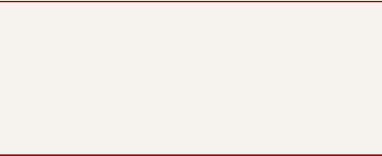
USB-MIDI-controlled volume and pan for two balanced 1/4" line channels. Bus powered, no external supply.
Sheets: 1 input, 2 matrix, 3 output, 4 control, 5 power. Nets between sheets are global labels; rails are power symbols.

Balanced inputs



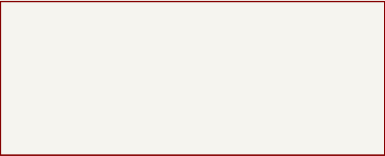
File: input.kicad_sch

2x2 gain matrix and summers



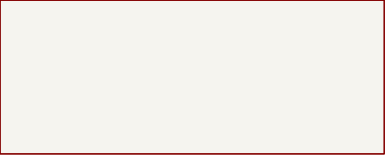
File: matrix.kicad_sch

Balanced outputs



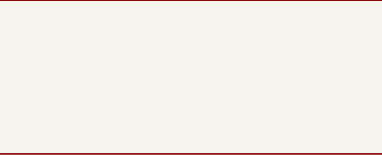
File: output.kicad_sch

Control



File: control.kicad_sch

Power



File: power.kicad_sch

Signal flow: J1/J2 -> U1/U2 THAT1246 (-6 dB) -> C3-C6 -> U3/U4 PGA2310 2x2 matrix -> R2-R9 + U5 OPA2134 summers (x2) -> U6/U7 THAT1646 (+6 dB) -> J3/J4
Control: Teensy 4.0 USB MIDI -> SPI (SCLK, SDI, /CS) + /MUTE -> U3 -> U4. Plug sense from J1/J2 ring-normal contacts.
Power: Teensy VIN (USB 5 V) -> FB1 -> PS1 TMR 3-0523 -> +/-15 V. Same 5 V feeds PGA logic.

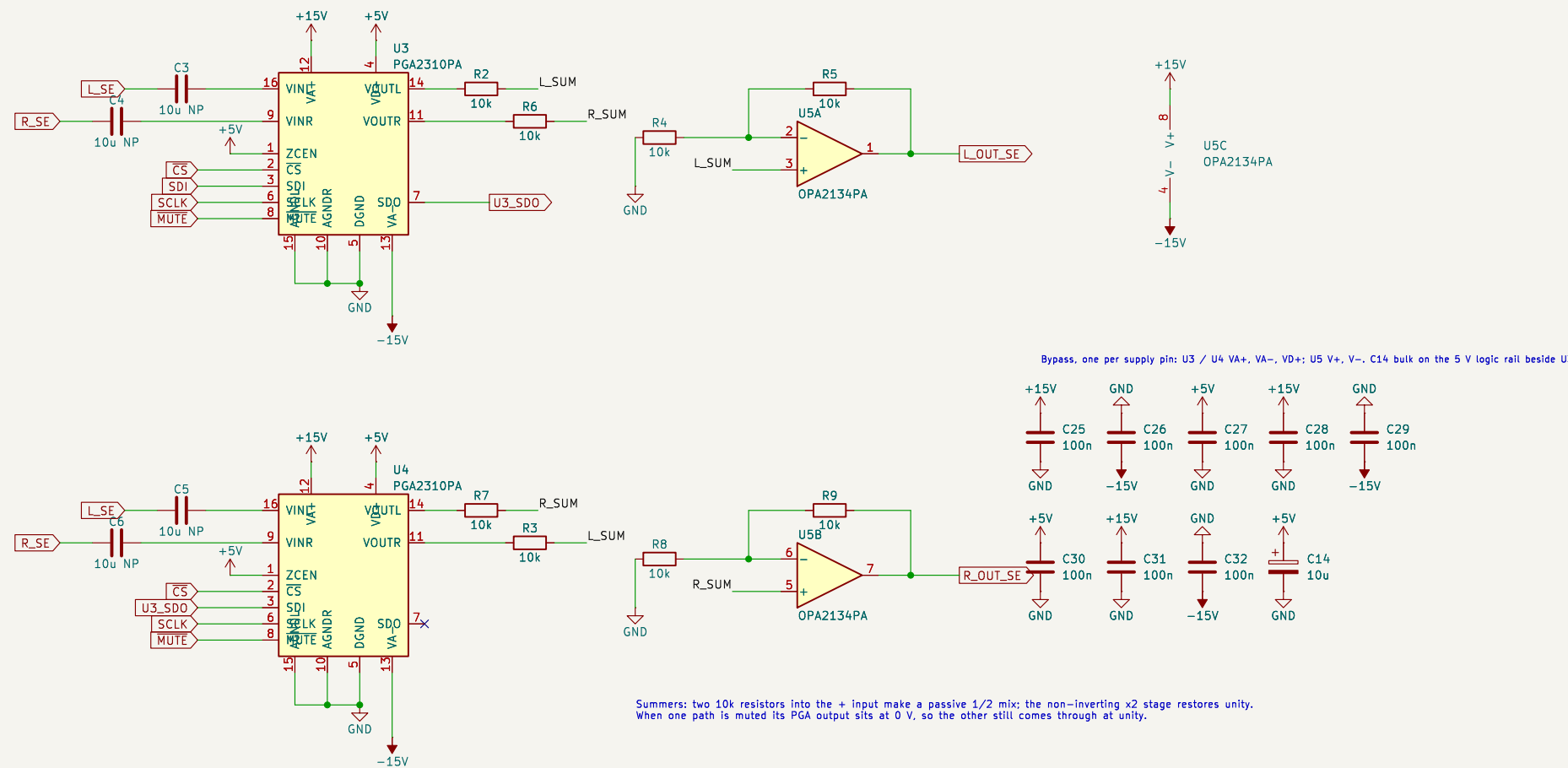
<https://chris-albert.github.io/ghost-fader/>
USB-MIDI volume and pan for two balanced line channels
Ghost Fader

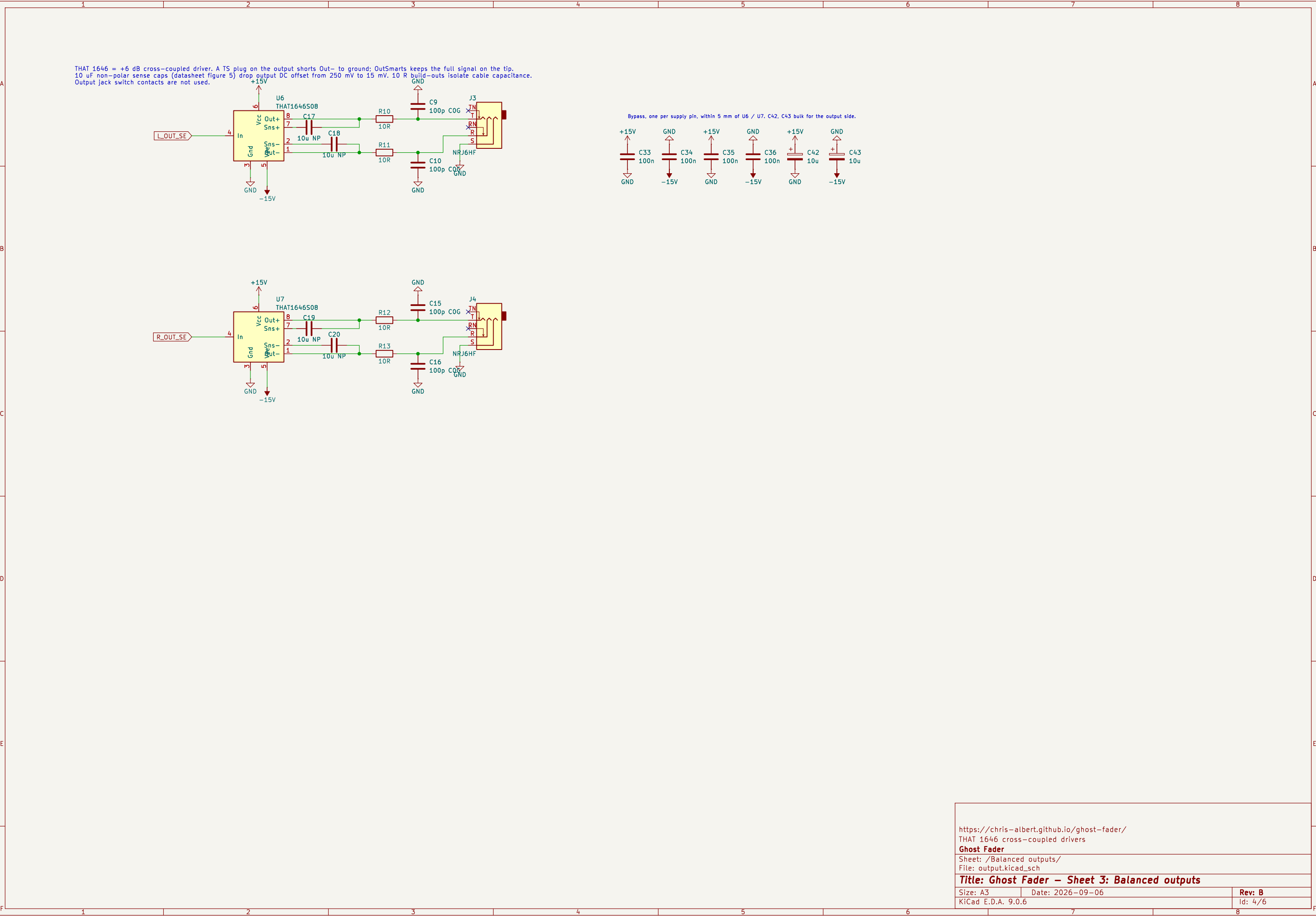
Sheet: /
File: ghost-fader.kicad_sch

Title: Ghost Fader – Rev B

Size: A4	Date: 2026-09-06	Rev: B
KiCad E.D.A. 9.0.6	Id: 1/6	

Size: A3	Date: 2026-09-06	Rev: B
KiCad E.D.A. 9.0.6		Id: 2/6





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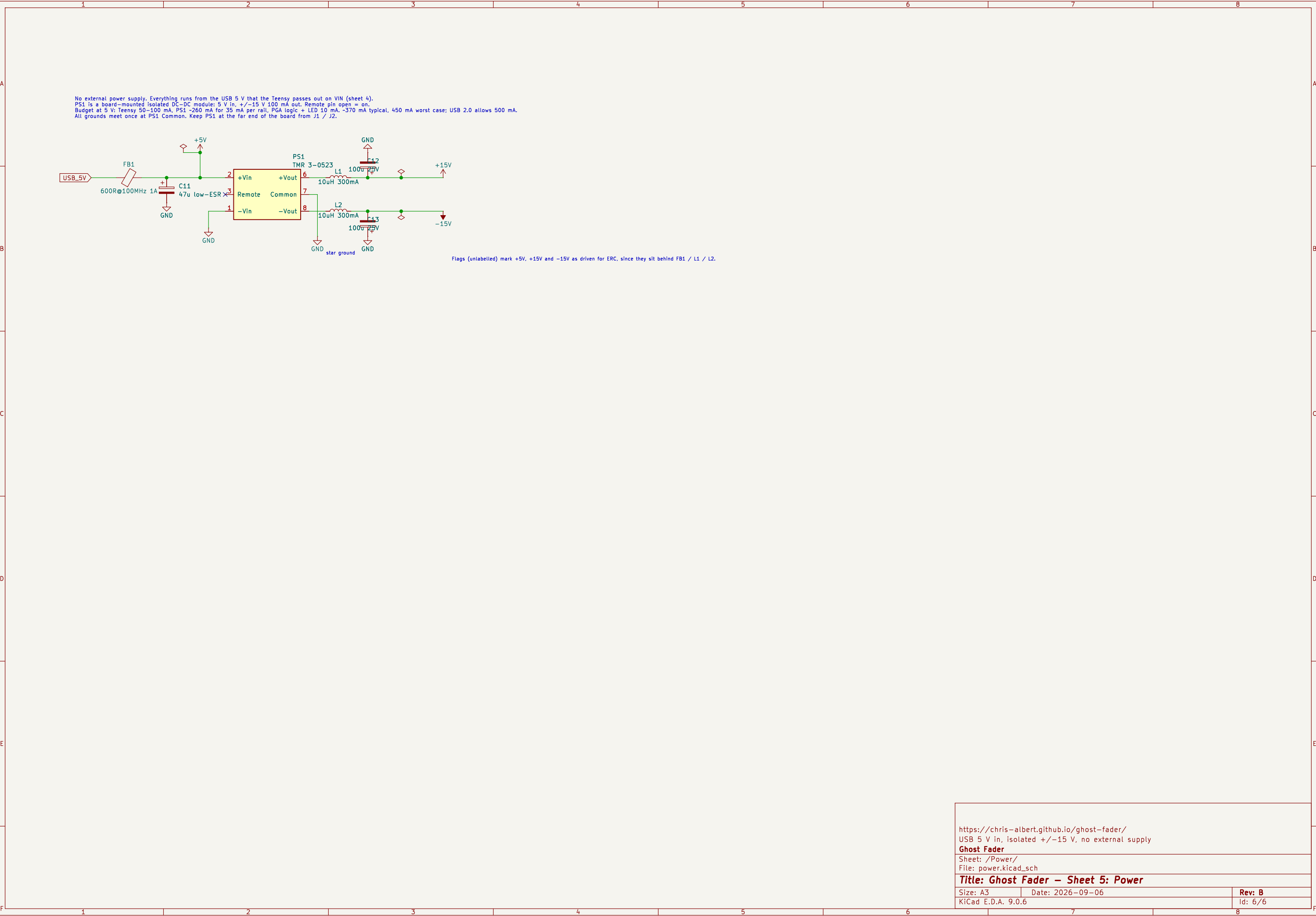
5

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1

1

KiCad E.D.A. 9.0.6



Flags (unlabelled) mark +5V, +15V and -15V as driven for ERC, since they sit behind FB1 / L1 / L2.

<https://chris-albert.github.io/ghost-fader/>
USB 5 V in, isolated +/-15 V, no external supply
Ghost Fader
Sheet: /Power/
File: power.kicad_sch

Title: Ghost Fader – Sheet 5: Power

Size: A3	Date: 2026-09-06	Rev: B
KiCad E.D.A. 9.0.6	Id: 6/6	